OCR - Based Contract Analysis and Simplification System

Sunkari Sai Varshini
Dept. of CSE
MALLA REDDY UNIVERSITY,
Hyderabad, India
saivarshini2709@gmail.com

Vaishali Khandoba
Dept. of CSE
MALLA REDDY UNIVERSITY,
Hyderabad, India
vaishali.khandoba7@gmail.com

Srimanthula Vamshi Kumar
Dept. of CSE
MALLA REDDY UNIVERSITY,
Hyderabad, India
vamshi14908@gmail.com

Yarlanki Uday Kiran
Dept. of CSE
MALLA REDDY UNIVERSITY,
Hyderabad, India
udaykiranuday750@gmail.com

GuideName : Mrs.. P.Sabitha
Assit prof, Dept. of CSE
MALLA REDDY UNIVERSITY,
Hyderabad, India

ABSTRACT

Legal contracts are fundamental instruments governing business, employment, finance, healthcare, and service transactions. However, contracts are often drafted using complex legal terminology and lengthy clauses that are difficult for non-legal users to understand. Manual contract review requires significant time, expertise, and financial resources, increasing operational costs and risk of oversight.

This paper presents an OCR-Based Contract Analysis and Simplification System, a lightweight web-based application that automates the extraction and analysis of contract documents. The system utilizes Optical Character Recognition (OCR) to convert scanned contract images into machine-readable text and applies rule-based Natural Language Processing (NLP) techniques for clause detection, sector classification, risk identification, summary generation, and legal term simplification.

The proposed solution integrates Tesseract.js for OCR processing and JavaScript-based text analysis to identify payment terms, termination clauses, liabilities, and confidentiality provisions. A heuristic risk scoring mechanism evaluates predefined risk- related keywords. The system further enhances accessibility by translating complex legal terminology into simplified plain English explanations.

Keywords - OCR, Natural Language Processing, Contract Analysis, LegalTech, Risk Detection, Text Simplification.

1 INTRODUCTION

Legal contracts define obligations, rights, liabilities, and responsibilities between parties. They are widely used in employment agreements, financial transactions, healthcare services, and corporate governance. Despite their importance, contracts are often written in complex legal language containing technical jargon, ambiguous clauses, and lengthy sentence structures.

Many individuals sign agreements without fully understanding their implications due to lack of legal expertise. Manual contract review is time-consuming and costly, especially for startups and small businesses. The emergence of Artificial Intelligence (AI) and Natural Language Processing (NLP) enables automated analysis of unstructured legal documents.

This research proposes an OCR-Based Contract Analysis and Simplification System that combines OCR and rule-based NLP techniques to automate contract understanding. The system extracts text from scanned contracts and analyzes it to identify key clauses, risk indicators, sector classification, and simplified explanations.

2 LITERATURE SURVEY

Document digitization has been widely enabled by OCR technology. The Tesseract OCR engine introduced by Smith [1] demonstrated high-accuracy character recognition in printed documents. OCR tools have since been integrated into document analysis systems.

NLP techniques such as tokenization, keyword extraction,

document classification are commonly used in legal document processing [4]. Research in legal analytics explores predictive modeling and machine learning approaches for clause detection and risk assessment [7][8]. However, these approaches require large annotated datasets and significant computational resources.

Rule-based systems remain effective in domain-specific applications due to their interpretability and efficiency. Legal text simplification techniques also aim to improve accessibility for non-expert users.

The proposed system builds upon OCR and rule-based NLP frameworks to create a lightweight and practical contract analysis solution.

Legal text simplification has also been studied to improve accessibility for non-expert users. Lexical simplification techniques replace complex legal terminology with plain language equivalents, enhancing readability and transparency.

Based on the reviewed literature, integrating OCR with rule-based text processing provides a practical and efficient solution for automated contract analysis. The proposed system builds upon these established concepts by combining OCR-based text extraction, keyword-driven classification, risk detection, clause extraction, and legal term simplification within a unified web- based framework.

3 SYSTEM ARCHITECTURE

The system consists of the following modules:

1. Document Upload Module
2. OCR Text Extraction Module
3. Text Preprocessing Module
4. Clause Detection Module
5. Risk Detection Module
6. Sector Classification Module
7. Summary and Key Point Extraction Module
8. Legal Term Simplification Module
9. PDF Report Generation Module

4 PROPOSED METHODODLOGY

The system follows a structured NLP pipeline:
Step 1: Document Upload
User uploads contract document (PDF or text file).

Step 2: Text Extraction
The system extracts raw text from the document.

Step 3: Text Preprocessing
• Convert to lowercase
• Remove punctuation
• Remove stop words
• Tokenize sentences and words
• Perform basic normalization

Step 4: Keyword Extraction
Important legal terms are identified using keyword matching.

Step 5: Clause Identification
The system scans for predefined clause indicators such as:
• Payment terms
• Termination clauses
• Liability clauses
• Confidentiality clauses

Step 6: Risk Detection
Risk-related terms are identified, such as:
• "Penalty"
• "Unlimited liability"
• "Breach"
• "Termination without notice"

Step 7: Risk Scoring
Risk Score = (Number of Risk Keywords × Assigned Weight) / Total Clauses

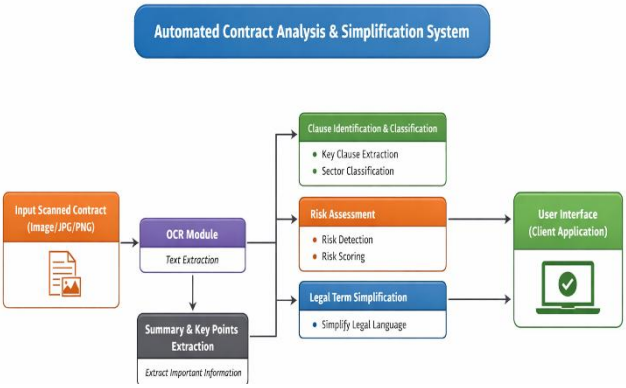
Step 8: Sector Classification
Based on dominant keywords, contract is classified into sectors like:
• Finance
• Employment
• Services
• Real Estate

Step 9: Summary Generation
Important sentences are extracted to form a concise summary.

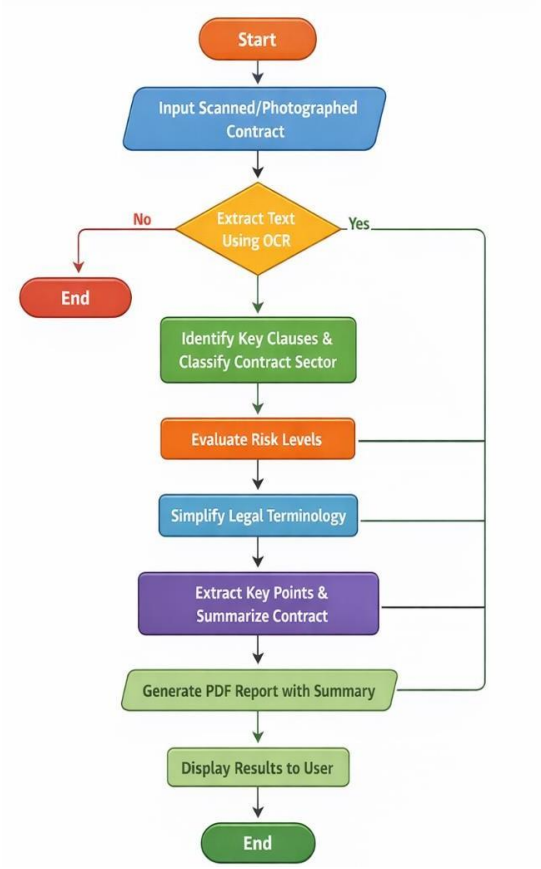
Step 10: Simplification
Complex legal terms are replaced with simpler alternatives.

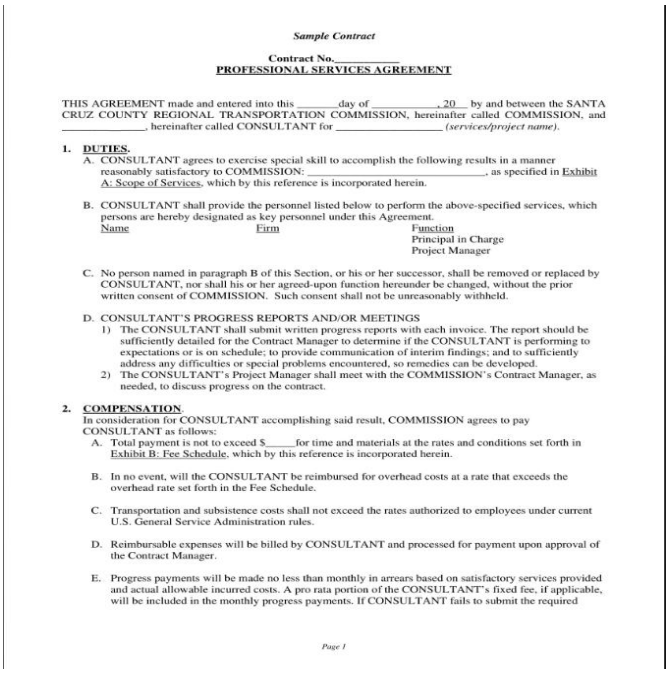
Step 11: Result Display
All outputs are shown in dashboard format.

5 BLOCK DIAGRAM



6 FLOW CHART





Sample Contract

Contract No._____
PROFESSIONAL SERVICES AGREEMENT

THIS AGREEMENT made and entered into this _____day of _____,20__ by and between the SANTA CRUZ COUNTY REGIONAL TRANSPORTATION COMMISSION, hereinafter called COMMISSION, and _____, hereinafter called CONSULTANT for _____ *(services/project name)*.

1. **DUTIES.**
   A. CONSULTANT agrees to exercise special skill to accomplish the following results in a manner reasonably satisfactory to COMMISSION: _____, as specified in Exhibit A: Scope of Services, which by this reference is incorporated herein.
   B. CONSULTANT shall provide the personnel listed below to perform the above-specified services, which persons are hereby designated as key personnel under this Agreement.
   Name Firm Function
   Principal in Charge
   Project Manager
   C. No person named in paragraph B of this Section, or his or her successor, shall be removed or replaced by CONSULTANT, nor shall his or her agreed-upon function hereunder be changed, without the prior written consent of COMMISSION. Such consent shall not be unreasonably withheld.
   D. CONSULTANT'S PROGRESS REPORTS AND/OR MEETINGS
   1) The CONSULTANT shall submit written progress reports with each invoice. The report should be sufficiently detailed for the Contract Manager to determine if the CONSULTANT is performing to expectations or is on schedule; to provide communication of interim findings; and to sufficiently address any difficulties or special problems encountered, so remedies can be developed.
   2) The CONSULTANT's Project Manager shall meet with the COMMISSION's Contract Manager, as needed, to discuss progress on the contract.
2. **COMPENSATION.**
   In consideration for CONSULTANT accomplishing said result, COMMISSION agrees to pay CONSULTANT as follows:
   A. Total payment is not to exceed $_____for time and materials at the rates and conditions set forth in Exhibit B: Fee Schedule, which by this reference is incorporated herein.
   B. In no event, will the CONSULTANT be reimbursed for overhead costs at a rate that exceeds the overhead rate set forth in the Fee Schedule.
   C. Transportation and subsistence costs shall not exceed the rates authorized to employees under current U.S. General Service Administration rules.
   D. Reimbursable expenses will be billed by CONSULTANT and processed for payment upon approval of the Contract Manager.
   E. Progress payments will be made no less than monthly in arrears based on satisfactory services provided and actual allowable incurred costs. A pro rata portion of the CONSULTANT's fixed fee, if applicable, will be included in the monthly progress payments. If CONSULTANT fails to submit the required

Page 1

Figure 7.1 Original Contract Paper

7 RESULTS

The OCR-Based Contract Analysis and Simplification System successfully performs automated extraction and analysis of contract documents. The system accepts a scanned contract image (JPG/PNG format) as input and uses
Tesseract.js to extract machine-readable text from the document. The OCR module demonstrates reliable performance when processing clear and well-formatted contract images, accurately recognizing the majority of visible text content.

After text extraction, the system applies rule-based keyword detection to identify important contractual terms such as liability, termination, penalty, indemnity, and jurisdiction. These keywords are mapped to predefined dictionary meanings to ensure clarity and transparency. The system also classifies the contract sector based on domain-specific keywords and determines a risk level using keyword-based analysis. Identified clauses and key sentences are extracted and presented in a structured format for easy understanding.

The final output is displayed through a user-friendly dashboard that includes the contract sector, risk level, summary, key points, and simplified legal terms. Additionally, the system generates a structured PDF report containing the analyzed results. The effectiveness of the system is evaluated based on OCR accuracy, correctness of keyword detection, clarity of simplified explanations, and overall usability. The results demonstrate that the proposed approach provides a practical and efficient solution for simplifying and analyzing contract documents using lightweight web technologies.

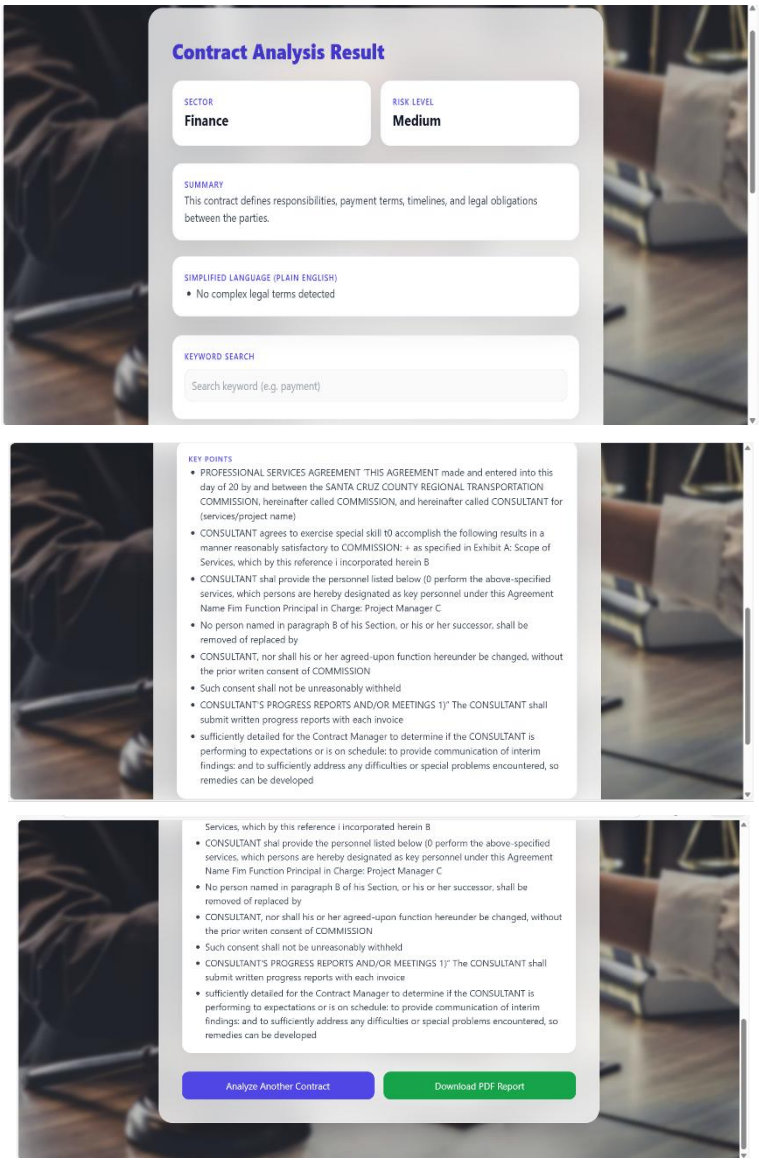


Figure 7.2 Interpreted Output File

8 CONCLUSION

This research presents an OCR-Based Contract Analysis and Simplification System designed to automate the extraction and analysis of legal contract documents. The system effectively converts scanned contract images into machine-readable text using Optical Character Recognition and applies rule-based text processing techniques to identify key clauses, detect risk-related terms, classify contract sectors, and simplify complex legal terminology. The results demonstrate that the proposed approach provides accurate text extraction and meaningful keyword-based interpretation without relying on machine learning models. The lightweight web-based implementation ensures accessibility, efficiency, and ease of use while reducing manual effort involved in contract review. Although the system is dependent on image quality and predefined keyword rules, it successfully illustrates the practical application of OCR and AI techniques in legal document automation and simplified contract understanding.

9 FUTURE SCOPE

- Advanced Risk Scoring Mechanism
- Machine Learning-Based Clause Classification
- Multi-Format Document Support (PDF, DOCX)
- Cloud-Based Contract Storage
- Multilingual Contract Analysis
- Real-Time Risk Alerts
- Dynamic Legal Dictionary Updates
- Integration with Legal Management Systems

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